

LLM Finetuning training loop

(1) Fine-tuning the model (step 5)

(a) Loss function: Cross entropy loss (similar to the one used for pre-training the LLM)

(b) Run training loop on instruction dataset

* Training loop and loss function are exactly the same as the training loop and loss function which we had looked at when we pre-trained model

Batch 1

1 → input

target

some value

predicted value

2 →

⋮

8 →

Loss Function

Cross entropy loss

* Due to pre-training, model starts from knowledgeable state, instead of a random state

↳ model is able to understand the semantic meaning of words
↳ when given a text, it knows how to respond.

↳ knows a lot about the language itself

↳ All Through Pre-training
↳ Thus, we require less data to fine-tune, b/c LLM already has so much background information

GPT Architecture

Losses

Probabilities